



EXPERIMENT NO-10

Realization of full adder using a 4 to 1 multiplexer and basic gates.



By Prof. Tanmoy Biswas, Syamaprasad College

BY: PROF. TANMOY BISWAS
Department of Computer Science, Syamaprasad College.

Experiment No – 10, Study and realization of a Full Adder using Dual 4 to 1 multiplexer 74153 and other necessary logic gates.

Theory:

A full adder is a digital circuit that performs addition on three binary inputs and produces two binary outputs. It's a fundamental building block in digital electronics and is used to add binary numbers. The three inputs to a full adder are typically labelled A, B, and C_{in} (carry-in), and the outputs are usually **Sum** (the sum of A, B, and C_{in}) and C_{out} (carry-out, which represents any carry generated by the addition).

C_{in}	B	A	Sum (S)	Carry (C_{out})
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 10.1, The truth table for a full adder.

The Sum output gives the result of adding A, B, and C_{in} , while the C_{out} output indicates if there's a carry generated, which is useful for chaining multiple full adders together to add multi-bit numbers.

Full adders can be used in combination to create adders for multi-bit binary numbers. A 4-bit binary adder, for example, would use four full adders chained together, with the C_{out} from one full adder connected to the C_{in} of the next full adder, and so on. This is how computers perform binary addition.

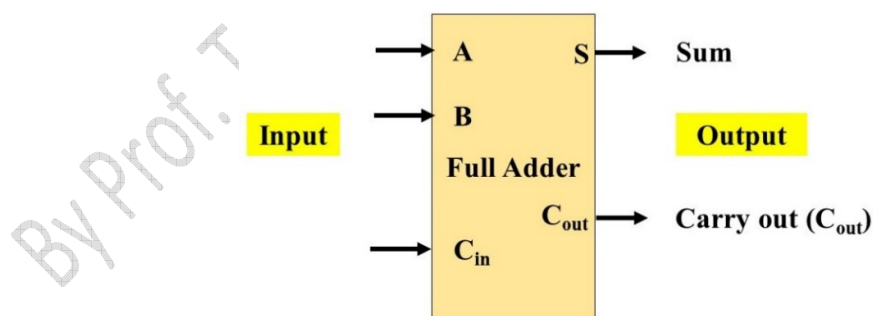


Figure (10.2), block diagram of full adder.

The SUM and CARRY outputs of a full adder can be expressed using Boolean logic. The Boolean expressions for SUM (S) and CARRY-OUT (C_{out}) in terms of the inputs A, B, and C_{in} are as follows:

SUM (S) expression:

$$S = \overline{C_{in}} \cdot \overline{B} \cdot A + \overline{C_{in}} \cdot B \cdot \overline{A} + C_{in} \cdot \overline{B} \cdot \overline{A} + C_{in} \cdot B \cdot A \dots\dots\dots (1)$$

Similarly, we can write the Boolean expression for the Carry also as given below;

$$C_{out} = \overline{C_{in}} \cdot B \cdot A + C_{in} \cdot \overline{B} \cdot A + C_{in} \cdot B \cdot \overline{A} + C_{in} \cdot B \cdot A \dots\dots\dots (2)$$

Realization of Full Adder using Dual 4 to 1 Multiplexer (74153).

Let us now consider the expression (1) and consider the input variable of Full adder B as S_1 and A as S_0 of the multiplexer select lines. Now let's compare the expression (1) with the characteristic expression of multiplexer as given in expression (3).

$$Y = I_0 \cdot (\overline{S_1} \cdot \overline{S_0}) + I_1 \cdot (\overline{S_1} \cdot S_0) + I_2 \cdot (S_1 \cdot \overline{S_0}) + I_3 \cdot (S_1 \cdot S_0) \dots\dots\dots (3)$$

Apply bracket to all the minterm with the variable B and A in the expression (1) as shown below;

$$S = \overline{C_{in}} \cdot (\overline{B} \cdot A) + \overline{C_{in}} \cdot (B \cdot \overline{A}) + C_{in} \cdot (\overline{B} \cdot \overline{A}) + C_{in} \cdot (B \cdot A) \dots\dots\dots (1)$$

Now let's compare the equations (1) and (3), if $B = S_1$ & $A = S_0$ then; I_0 can be equated to $\overline{C_{in}}$, similarly, I_1 can be equated to $\overline{C_{in}}$, $I_2 = \overline{C_{in}}$, and $I_3 = C_{in}$.

Similarly, let us consider the expression for Carry (C_{out}), and bracket the variables or the minterm with B and A.

$$C_{out} = \overline{C_{in}} \cdot (B \cdot A) + C_{in} \cdot (\overline{B} \cdot A) + C_{in} \cdot (B \cdot \overline{A}) + C_{in} \cdot (B \cdot A)$$

$$C_{out} = (\overline{C_{in}} + C_{in}) \cdot (B \cdot A) + C_{in} \cdot (\overline{B} \cdot A) + C_{in} \cdot (B \cdot \overline{A}) + 0 \cdot (\overline{B} \cdot \overline{A})$$

We can see in the equation (4) that after taking common we are left with the following as shown in the equation (4).

$$C_{out} = 1 \cdot (B \cdot A) + C_{in} \cdot (\overline{B} \cdot A) + C_{in} \cdot (B \cdot \overline{A}) + 0 \cdot (\overline{B} \cdot \overline{A}) \dots\dots\dots (4)$$

We are also adding the fourth minterm where it is ANDed with zero. Now, let us compare the equation (4) with the expression of the multiplexer.

$$Y = I_0 \cdot (\overline{S_1} \cdot \overline{S_0}) + I_1 \cdot (\overline{S_1} \cdot S_0) + I_2 \cdot (S_1 \cdot \overline{S_0}) + I_3 \cdot (S_1 \cdot S_0)$$

Now let's compare the equations (4) and multiplexer characteristics equation, if $B = S_1$ & $A = S_0$ then; I_0 can be equated to 1, similarly, I_1 & I_2 can be equated to C_{in} , and $I_3 = C_{in}$.

Now let's realize the Full Adder using dual 4 to 1 multiplexer (74153).

The figure (10.3) shows the realization of full adder.

Implementation and Circuit diagram

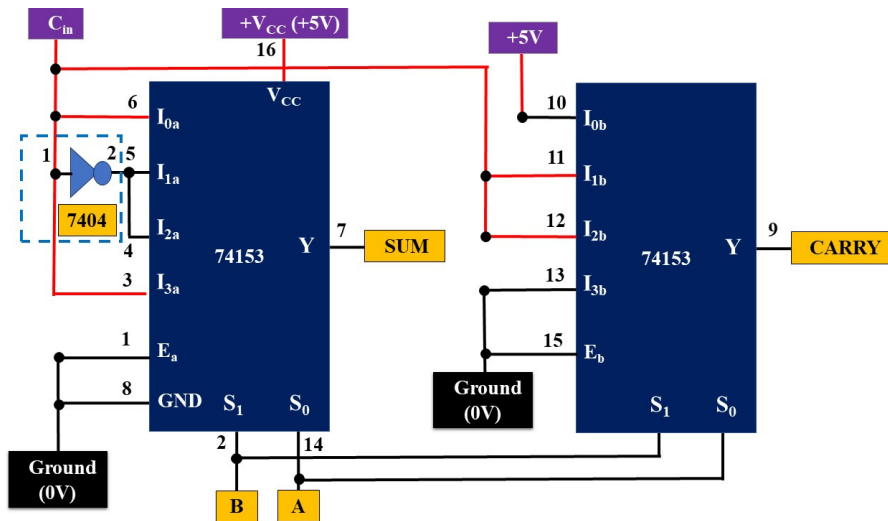


Figure (10.3) shows the realization of full adder using dual 4 to 1 multiplexer.

Semiconductors and Instruments required

Power Supply

1. DC +5V regulated power supply.

Integrated Circuits

1. Dual 4 to 1 multiplexer – 74153.
2. Hex Inverter – 7404.

Measuring Scientific equipment's

1. Digital Multimeter

Miscellaneous

1. Wish Board / Bread board
2. Jumper / connecting wires

Result/Observation

Truth table for the full adder using dual 4 to 1 multiplexer and basic logic gates.

Input			Logical Output		Output Voltage	
C	B	A	SUM	CARRY	SUM	CARRY
0	0	0	0	0		
0	0	1	1	0		
0	1	0	1	0		
0	1	1	0	1		
1	0	0	1	0		
1	0	1	0	1		
1	1	0	0	1		
1	1	1	1	1		

Table (10.2), Experimental observation of realization of full adder using 4 to 1 multiplexer and basic gates.

Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V** (V_{CC}) or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.
6. In this circuit we are realizing a half adder using a dual 4 to 1 multiplexer 74153.
7. To enable the multiplexer, we need to ground or connect the pin number 1 and 15.
8. The select lines S_1 and S_0 are assumed to be the full adder inputs A and B. Both the multiplexer inside the 74153 has common select lines.
9. We must connect a DC regulated power supply, outputting +5 volts, to pin 16 and ground to pin 8 in order to provide the necessary supply voltage for the circuit.